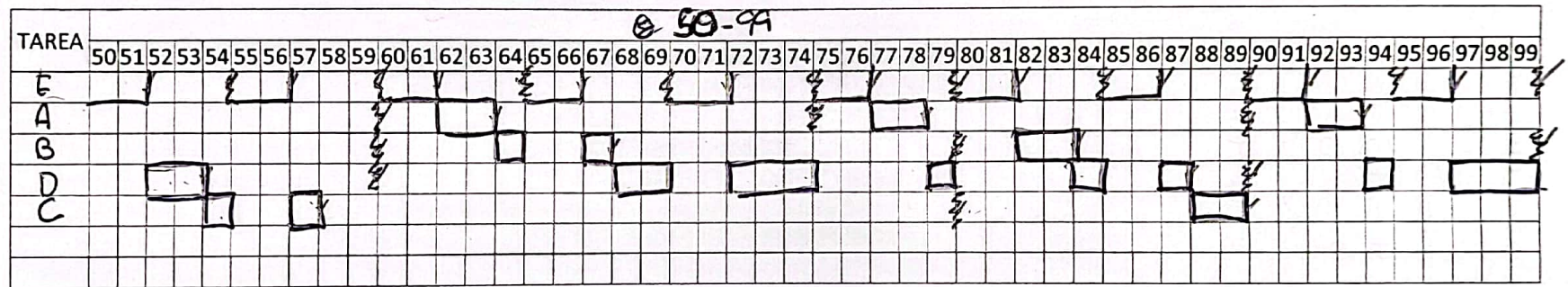
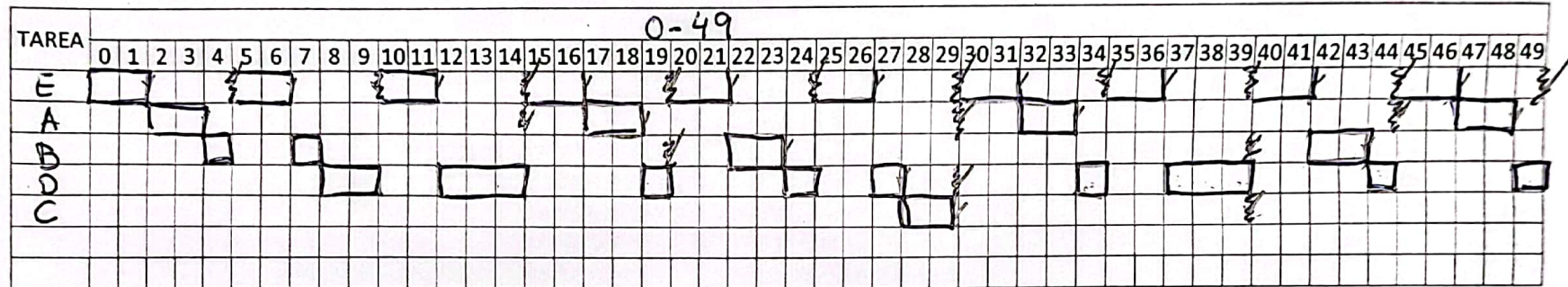


Tablas en blanco para poder usar en los problemas



Tarea	Periodo (P)	Tiempo Ejecución (C)	Prioridad	Grado Utilización
A	15	2	4	$2/15 = 0.13$
B	20	2	3	$2/20 = 0.1$
C	40	2	1	$2/40 = 0.05$
D	30	8	2	$8/30 = 0.26$
E	5	2	5	$2/5 = 0.4$
Grado utilización total:				$19/20 = 0.95$

menor periodo $\rightarrow$ Mayor prioridad ($C_i < T_j \rightarrow P_i > P_j$)
 Grado de Utilización $\rightarrow C_i/T_i$

El mayor $\rightarrow$ MCM (Periodos) = 120

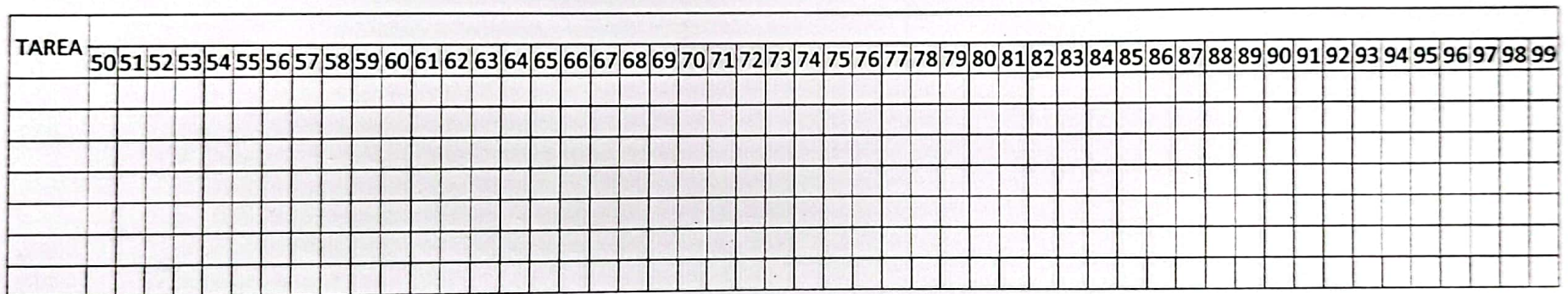
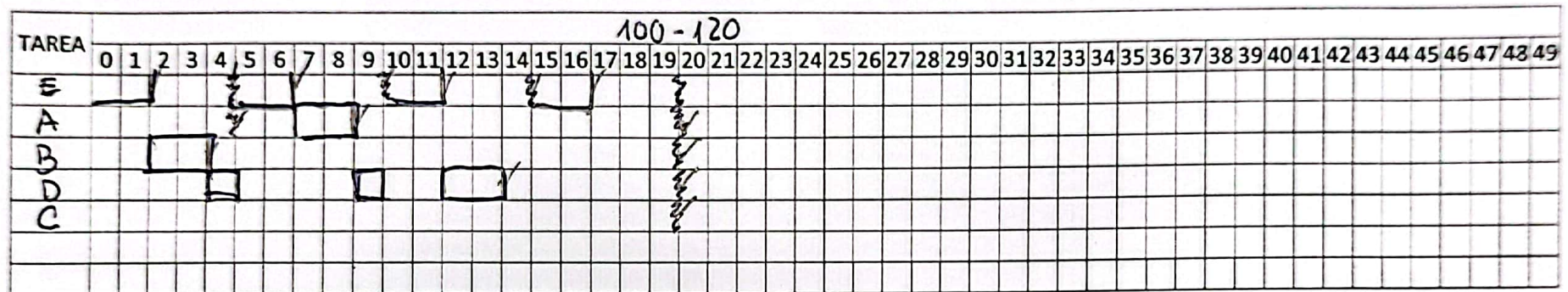
a) Test de planificabilidad (Lui y Layland)

$$\sum_{i=1}^N \left(\frac{C_i}{T_i} \right) < N \cdot (2^{1/N} - 1) \quad N = 5$$

$$0.95 < 0.7435$$

$\rightarrow$ No se cumple = NO SABEMOS SI ES PLANIFICABLE.

Tablas en blanco para poder usar en los problemas



Tarea	Periodo (P)	Tiempo Ejecución (C)	Prioridad	Grado Utilización
Grado utilización total:				

b) Test de Tiempo de Respuesta de RNA.

$$R_i = C_i + \sum_{j \in \text{mpa}(i)}^N \left[\frac{R_j}{T_j} \right] \cdot C_j$$

$$R_E = 2$$

$$\boxed{R_A}$$

$$R_A = 2 + \frac{R_A}{5} \cdot 2$$

$$\dot{R}_A = 3? \rightarrow 3 = 2 + \frac{3}{5} \cdot 2 ; 3 \neq 4$$

$$\dot{R}_A = 4? \rightarrow 4 = 2 + \frac{4}{5} \cdot 2 ; 4 = 4$$

$$\underline{\underline{R_A = 4}}$$

$$\boxed{R_B}$$

$$R_B = 2 + \left[\frac{R_B}{15} \cdot 2 + \frac{R_B}{5} \cdot 2 \right]$$

$$\dot{R}_B = 5? \rightarrow 5 = 2 + \left[\frac{5}{15} \cdot 2 + \frac{5}{5} \cdot 2 \right]$$

$$5 \neq 6$$

$$\dot{R}_B = 6? \rightarrow 6 = 2 + \left[\frac{6}{15} \cdot 2 + \frac{6}{5} \cdot 2 \right]$$

$$\underline{\underline{R_B = 8}}$$

$$6 \neq 8$$

$$\dot{R}_B = 8? \rightarrow 8 = 2 + \left[\frac{8}{15} \cdot 2 + \frac{8}{5} \cdot 2 \right]$$

$$8 = 8$$

$$\boxed{R_D}$$

$$R_D = 8 + \left[\frac{R_D}{20} \cdot 2 + \frac{R_D}{15} \cdot 2 + \frac{R_D}{5} \cdot 2 \right]$$

$$\dot{R}_D = 9? \rightarrow 9 = 8 + \left[\frac{9}{20} \cdot 2 + \frac{9}{15} \cdot 2 + \frac{9}{5} \cdot 2 \right] \neq$$

$$9 \neq 16$$

$$\dot{R}_D = 16? \rightarrow 16 = 8 + \left[\frac{16}{20} \cdot 2 + \frac{16}{15} \cdot 2 + \frac{16}{5} \cdot 2 \right]$$

$$16 = 8 + [2 + 4 + 8] \quad 16 \neq 22$$

$$\dot{d}R_D = 22? \quad 22 = 8 + \left[\frac{22}{20} \cdot 2 + \frac{22}{15} \cdot 2 + \frac{22}{5} \cdot 2 \right]$$

$$22 = 8 + [2 \cdot 2 + 4 + 10]$$

$$22 \neq 26$$

$$\dot{d}R_D = 26? \quad 26 = 8 + \left[\frac{26}{20} \cdot 2 + \frac{26}{15} \cdot 2 + \frac{26}{5} \cdot 2 \right]$$

$$26 = 8 + [4 + 4 + 12]$$

$$26 \neq 28$$

$$\dot{d}R_D = 28? \quad 28 = 8 + \left[\frac{28}{20} \cdot 2 + \frac{28}{15} \cdot 2 + \frac{28}{5} \cdot 2 \right]$$

$$28 = 8 + [4 + 4 + 12]$$

$$28 = 28$$

$$\underline{\underline{R_D = 28}}$$

R_C

$$R_C = 2 + \left[\frac{R_C}{30} \cdot 8 + \frac{R_C}{20} \cdot 2 + \frac{R_C}{15} \cdot 2 + \frac{R_C}{5} \cdot 2 \right]$$

$$\dot{d}R_C = 29? \quad 29 = 2 + \left[\frac{29}{30} \cdot 8 + \frac{29}{20} \cdot 2 + \frac{29}{15} \cdot 2 + \frac{29}{5} \cdot 2 \right]$$

$$29 = 2 + [8 + 4 + 4 + 12]$$

$$29 \neq 30$$

$$\dot{d}R_C = 30? \quad 30 = 2 + \left[\frac{30}{30} \cdot 8 + \frac{30}{20} \cdot 2 + \frac{30}{15} \cdot 2 + \frac{30}{5} \cdot 2 \right]$$

$$30 = 2 + [8 + 4 + 4 + 12]$$

$$30 = 30$$

$$\underline{\underline{R_C = 30}}$$